Oct 20, 2025

Meeting Oct 20, 2025 at 15:45 CEST

Meeting records Recording

Summary

Clément Lacaille led week three by introducing Telegram voice agents, outlining their functionality for processing voice messages, transcribing them, and responding in a synthesized voice using N8N, 11 Labs, and potentially vector databases. Ahmad Naweed Zahed volunteered to lead the collaborative workflow development and successfully created a Telegram bot using BotFather, with guidance from Clément. Participants including Murphy Odion Usunobun and Sara Hofmann troubleshooted Telegram token and login issues, while Clément demonstrated how to retrieve a file ID from a voice message and integrate ElevenLabs for transcription, emphasizing its voice cloning and text-to-speech features.

Details

Notes Length: Standard

- **Learning Goals and Feedback** Clément Lacaille welcomed everyone to week three, acknowledging their commitment and shared goal of learning new skills for career advancement. Clément also sought feedback on the previous week's format and content to tailor the class effectively, and Naim Burrack suggested incorporating a short five-minute break during sessions due to the intensity of the previous week. Ahmad Naweed Zahed praised the previous week's content but reported losing their workflow, potentially due to accidental deletion or not properly naming it, prompting Clément to advise saving workflows daily and exploring restoration options.

- **Telegram Voice Agents** Clément introduced voice agents as a key topic for the week, highlighting their increasing demand in various fields like HR, marketing, and programming, and noted that tools like "level labs" make voice agents highly conversational and human-like. Clément explained that the primary goal for the week is to build a voice agent that can process voice messages from Telegram, transcribe them, and respond in a synthesized voice. This will be achieved using N8N, a low-code tool, and other integrated applications like 11 Labs for voice processing, and potentially a vector database for information retrieval.
- **Voice Agent Functionality** Clément detailed the functionality of the voice agent, explaining that users would send a voice message via Telegram, which would then be transcribed by 11 Labs, processed by an agent, and replied to in the user's cloned voice. The agent could search online using a tool like PublicAI or retrieve information from a custom database, such as a Google document, to answer questions or provide data, making it useful for various applications like finding restaurants, job searching, or providing educational assistance. Clément emphasized the agent's ability to act as a "personal Jarvis or Alexa" that can even be trained to speak in the user's cloned voice by sending a 10-20 second audio sample.
- **Introduction to Vector Databases** Clément also mentioned the concept of a vector database, which handles unlabeled data, differentiating it from traditional structured databases like Excel. Clément explained that while complex, a vector database could enable agents to "remember" user preferences and past interactions, making them more intelligent and personalized, similar to a Spotify recommendation engine. Clément plans to delve deeper into vector databases, including how to create one using Pinecone, if the class progresses at a good pace.
- **Collaborative Workflow Development** Clément encouraged a student to lead the session by sharing their screen and guiding the class through the process of setting up the voice agent workflow. Ahmad Naweed Zahed volunteered to take the lead, and Clément guided them in starting a new workflow in PublicAI, providing the initial "role, context, and task" prompts for the AI to generate step-by-step instructions for building the Telegram voice bot.
- **Telegram Desktop Setup** Clément instructed all participants to download and install the Telegram desktop application, as it is crucial for the workflow and can be a valuable tool for creating front-end interfaces for agents. Sara Hofmann and Chinedu Kirian Onwuzuruike encountered issues logging into Telegram desktop,

with Sara asking about scanning a QR code with their mobile and Chinedu not receiving a code. Naim Burrack shared their successful method of scanning a QR code via the Telegram app's "link devices" option and suggested disabling two-step verification as a potential solution.

- **Telegram Bot Creation via BotFather** Ahmad Naweed Zahed, after successfully logging into Telegram desktop, was guided by Clément to search for "BotFather" within Telegram to create a new bot. Ahmad Naweed Zahed then named their bot "Navidid voice_assistant_bot," and upon successful creation, BotFather provided a unique token to access the HTTP API, which Clément instructed them to copy and save securely. Kai Feldhoff also followed the same process to create their bot and obtain the API token.
- **Personal Voice Assistant Setup** Clément Lacaille guided Ahmad Naweed Zahed through setting up their personal voice assistant, emphasizing the use of the bot for communication and to follow instructions from "publicity" for initial setup. Sara Hofmann encountered login issues, which Clément Lacaille addressed by providing access to an API keys document containing login credentials.
- **Workflow Triggers** Clément Lacaille explained that every workflow needs a starting point called a "trigger," which initiates the workflow. Ahmad Naweed Zahed confirmed they had named their workflow and proceeded to set up the second trigger.
- **Telegram Integration** To integrate Telegram, Clément Lacaille instructed Ahmad Naweed Zahed to add a Telegram trigger and select "on message" as the operation. They then guided Ahmad Naweed Zahed to enter their Telegram token into the edit icon next to the Telegram account.
- **Troubleshooting Token and Login Issues** Murphy Odion Usunobun faced issues with their Telegram token, which Clément Lacaille helped resolve by guiding them to create a new credential and paste their token, ensuring proper naming for clarity. Sara Hofmann also had a similar issue with credentials, and Clément Lacaille provided step-by-step instructions to create new credentials, paste the token, and name the account.
- **Voice Message Functionality Test** Clément Lacaille directed Ahmad Naweed Zahed to test the voice message functionality in Telegram after executing the workflow. Ahmad Naweed Zahed recorded a voice message, and Clément Lacaille provided guidance on how to properly record and stop the voice message, including dragging the voice button up.

- **Workflow Execution and Error Resolution** Ahmad Naweed Zahed encountered an issue where their workflow was not stopping or working, and Clément Lacaille suggested refreshing the page after saving the workflow. An error about "maximum cost tax executed" suggested potential token misuse or too many requests, prompting Clément Lacaille to advise sending a text message first to test the bot.
- **File ID Retrieval for Voice Messages** Clément Lacaille demonstrated how to retrieve a "file ID" from a voice message sent via Telegram, noting that sending a text message first might resolve issues with voice message recognition. Sara Hofmann successfully obtained a file ID after troubleshooting similar issues, which involved refreshing and re-executing the workflow.
- **Downloading Voice Files** Clément Lacaille instructed Ahmad Naweed Zahed to add a new Telegram node in N8N to download the voice file using the "get a file" operation and referencing the file ID from the previous Telegram trigger. They also showed an alternative method of dragging and dropping the file ID from the left-hand side panel.
- **ElevenLabs Integration and Transcription** Clément Lacaille guided Ahmad Naweed Zahed to add an ElevenLabs node to their workflow for transcribing audio and video. They introduced ElevenLabs as a powerful tool capable of creating voice content, cloning voices, and converting text to speech in multiple languages.
- **ElevenLabs Features and API Key** Clément Lacaille elaborated on ElevenLabs' features, including voice cloning from short audio samples and advanced text-to-speech models like Levelas V3. They also explained how to create an API key in ElevenLabs for future use, emphasizing its importance for accessing features.
- **Voice Agent Creation and Future Plans** Clément Lacaille demonstrated how to create a voice agent within ElevenLabs, highlighting its potential for personal assistants or learning companions and its integration with other platforms. They outlined plans to continue the session by focusing on transcribing audio from ElevenLabs and setting up an AI voice agent to generate responses.
- **Exploring Voice Technology Tools** Clément Lacaille encouraged participants to explore ElevenLabs and other tools like HeyGen, a video creation tool that can combine voice clones and avatars, to create content. They emphasized the rapid advancement and utility of these tools for content creation.

Suggested next steps

- [ ] Clément Lacaille will keep in mind to include a short 5-minute break during sessions.
- [ ] Naim Burrack will send Ahmad Naweed Zahed the finished workflow from Friday that used Ahmad Naweed Zahed's CV, which will require the least amount of changes.
- [ ] Clément Lacaille will investigate a better workflow management solution for working on the cloud with 30 people and explore ways to restore archived workflows.
- [ ] Clément Lacaille will share a video after the class showing an example of the Jarvis and Alexa-like voice agent they are creating.
- [ ] Naim Burrack will send Clément Lacaille a 20-second audio sample in MP3 format to get Naim Burrack's voice cloned.